

9. TEST ENVIRONMENT SETUP

A properly configured test environment is essential for successful execution of automation and security testing activities. The Test Environment Setup phase involves installing and configuring all software, tools, and applications required for the project.

For this project, the environment consists of Java 21, Eclipse IDE, Maven, XAMPP Server, DVWA Application, MySQL Database, Google Chrome Browser, and OWASP ZAP. These components work together to support automation testing, vulnerability testing, and security scanning activities.

9.1 Java 21 Installation

Java is the primary programming language used for developing the automation framework. Selenium WebDriver, TestNG, and Maven require Java to execute test scripts.

Installation Steps

1. Download Java 21 JDK from the Oracle website.
2. Run the installer and follow the installation wizard.
3. Install Java in the default directory.
4. Configure the JAVA_HOME environment variable.
5. Add the Java bin folder to the system PATH variable.

Verification

Open Command Prompt and execute:

```
java -version
```

Expected Output:

```
java version "21"
```

Purpose in Project

Java is used for:

- Selenium automation scripting
- TestNG test execution
- OWASP ZAP integration
- Report generation
- Framework development

Benefits

- Platform independent
 - Object-oriented programming support
 - Large community support
 - Strong integration with Selenium and TestNG
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9.2 Eclipse Setup

Eclipse IDE is used as the development environment for writing, editing, debugging, and executing automation scripts.

Installation Steps

1. Download Eclipse IDE for Java Developers.
2. Install Eclipse on the local machine.
3. Launch Eclipse.
4. Create a new workspace.
5. Configure Java 21 as the default JDK.

Project Configuration

Create a Maven Project and organize folders as follows:

```
src/test/java
src/main/java
src/test/resources
reports
screenshots
test-output
```

Purpose in Project

Eclipse is used for:

- Writing Selenium scripts
- Managing project files
- Debugging test scripts
- Executing TestNG tests
- Managing Maven projects

Benefits

- User-friendly interface
 - Integrated debugging tools
 - Maven support
 - TestNG plugin support
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9.3 Maven Setup

Maven is a build automation and dependency management tool used to manage project libraries and execute builds.

Installation Steps

1. Download Apache Maven.
2. Extract Maven files.
3. Configure MAVEN_HOME environment variable.
4. Add Maven bin folder to system PATH.
5. Verify installation.

Verification

Execute:

```
mvn -version
```

Expected Output:

```
Apache Maven 3.x.x  
Java Version: 21
```

Maven Dependencies Used

The project includes:

- Selenium WebDriver
- TestNG
- WebDriverManager
- Extent Reports
- OWASP ZAP API Client

Purpose in Project

Maven is used for:

- Dependency management
- Project build automation
- Test execution
- Report generation

Benefits

- Automatic dependency download
- Simplified project management
- Easy build execution

9.4 XAMPP Installation

XAMPP provides the local server environment required to host DVWA.

Components Included

- Apache Server
- MySQL Database
- PHP
- phpMyAdmin

Installation Steps

1. Download XAMPP.
2. Install XAMPP.
3. Launch XAMPP Control Panel.
4. Start Apache Service.
5. Start MySQL Service.

Verification

Open browser:

`http://localhost`

The XAMPP dashboard should be displayed successfully.

Purpose in Project

XAMPP is used to:

- Host DVWA locally
- Provide database connectivity
- Support PHP execution

Benefits

- Easy installation
- Lightweight setup
- Local testing environment

9.5 DVWA Configuration

DVWA (Damn Vulnerable Web Application) is the target application used for security testing.

Installation Steps

1. Download DVWA.
2. Extract files.
3. Copy DVWA folder into:

```
C:\xampp\htdocs\
```

4. Open browser:

```
http://localhost/DVWA
```

Configuration Steps

Open:

```
config/config.inc.php
```

Configure database credentials:

```
$_DVWA['db_server'] = '127.0.0.1';  
$_DVWA['db_database'] = 'dvwa';  
$_DVWA['db_user'] = 'root';  
$_DVWA['db_password'] = '';
```

Save the file.

Database Setup

Open:

```
http://localhost/DVWA/setup.php
```

Click:

```
Create / Reset Database
```

Security Level Configuration

Login using:

```
Username: admin  
Password: password
```

Navigate:

```
DVWA Security
```

Set Security Level:

Low

Purpose in Project

DVWA provides modules for:

- SQL Injection Testing
 - XSS Testing
 - Authentication Testing
 - Security Assessment
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9.6 MySQL Database Setup

MySQL stores DVWA application data and supports SQL Injection testing scenarios.

Setup Steps

1. Open XAMPP Control Panel.
2. Start MySQL Service.
3. Open phpMyAdmin:

`http://localhost/phpmyadmin`

4. Verify DVWA database creation.

Database Name:

`dvwa`

Verification

Check tables such as:

`users`
`guestbook`

Purpose in Project

MySQL supports:

- User authentication
- Vulnerability testing
- Database interaction

Benefits

- Open-source database
 - Easy integration with DVWA
 - Supports SQL Injection testing
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9.7 Chrome Browser Setup

Google Chrome is used as the primary browser for automation testing.

Installation Steps

1. Download Google Chrome.
2. Install browser.
3. Update to latest version.

ChromeDriver Setup

Download ChromeDriver matching browser version.

Store driver in project path or use WebDriverManager.

Example:

```
WebDriverManager.chromedriver().setup();
```

Verification

Launch browser through Selenium script.

Expected Result:

Chrome browser should open successfully.

Purpose in Project

Chrome is used for:

- Login automation
- SQL Injection testing
- XSS testing
- Security validation

Benefits

- Fast execution
- Stable browser automation
- Selenium compatibility

9.8 OWASP ZAP Configuration

OWASP ZAP (Zed Attack Proxy) is the primary security testing tool used in this project.

It performs:

- Spider Scans
- Active Scans
- Vulnerability Detection
- Security Reporting

Installation Steps

1. Download OWASP ZAP.
2. Install the application.
3. Launch OWASP ZAP.

API Configuration

Navigate:

Tools → Options → API

Configure:

Enable API = Checked
Disable API Key = Checked

OR

Generate API Key and update framework configuration.

Proxy Configuration

Default Settings:

Host: localhost
Port: 8080

Verification

Open browser:

`http://127.0.0.1:8080`

Or verify API access:

`http://127.0.0.1:8080/JSON/core/view/version/`

Selenium Integration

Project Configuration:

```
zap.proxy.host=localhost  
zap.proxy.port=8080  
zap.api.key=
```

Security Scan Setup

Configure target URL:

```
http://localhost/DVWA
```

Execute:

1. Spider Scan
2. Active Scan
3. Generate Security Report

Purpose in Project

OWASP ZAP is used to:

- Discover application pages
- Identify vulnerabilities
- Classify risks
- Generate security reports

Benefits

- Open-source security tool
 - OWASP recommended
 - Automation support
 - Comprehensive vulnerability scanning
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Environment Setup Summary

The test environment consists of the following components:

Component	Purpose
Java 21	Automation Development
Eclipse IDE	Script Development
Maven	Build & Dependency Management
XAMPP	Local Web Server
DVWA	Vulnerable Test Application
MySQL	Database Management
Chrome Browser	Test Execution
OWASP ZAP	Security Scanning

After successful completion of the environment setup, the system is ready for executing Login Testing, SQL Injection Testing, XSS Testing, OWASP ZAP Security Scanning, TestNG Execution, and Report Generation activities.